

Report
On
Terro's real estate agency

1) Generate the summary statistics for each variable in the table.

CRIME_RATE		AGE		INDUS	
Mean	4.871976285	Mean	68.57490119	Mean	11.13677866
Standard Error	0.129860152	Standard Error	1.251369525	Standard Error	0.304979888
Median	4.82	Median	77.5	Median	9.69
Mode	3.43	Mode	100	Mode	18.1
Standard Deviation	2.921131892	Standard Deviation	28.14886141	Standard Deviation	6.860352941
Sample Variance	8.533011532	Sample Variance	792.3583985	Sample Variance	47.06444247
	-		-		-
Kurtosis	1.189122464	Kurtosis	0.967715594	Kurtosis	1.233539601
Skewness	0.021728079	Skewness	-0.59896264	Skewness	0.295021568
Range	9.95	Range	97.1	Range	27.28
Minimum	0.04	Minimum	2.9	Minimum	0.46
Maximum	9.99	Maximum	100	Maximum	27.74
Sum	2465.22	Sum	34698.9	Sum	5635.21
Count	506	Count	506	Count	506

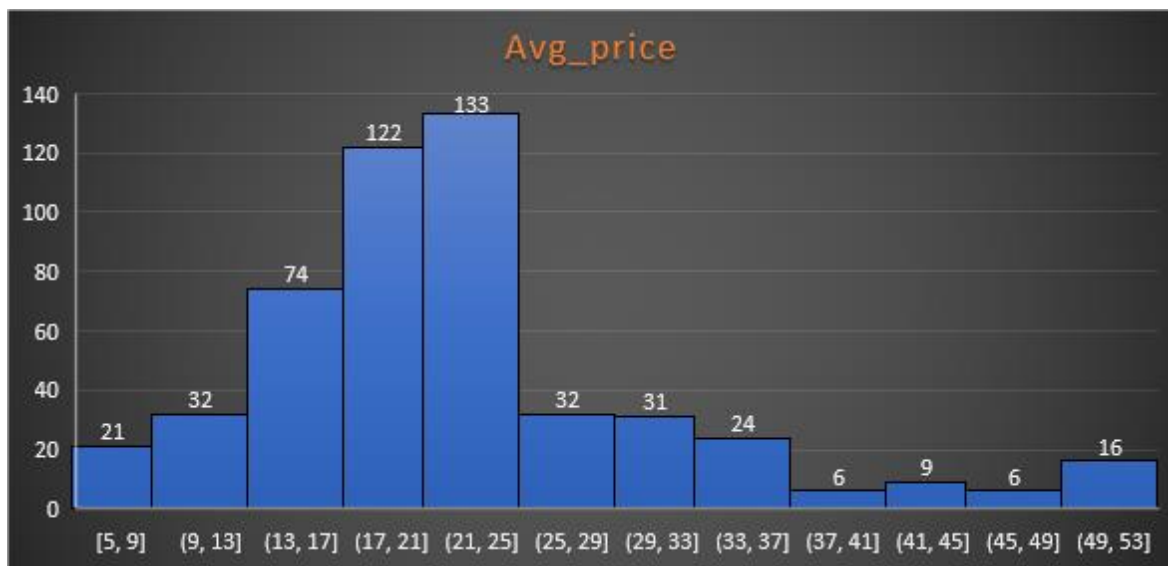
NOX		DISTANCE		TAX	
Mean	0.554695059	Mean	9.549407115	Mean	408.2371542
Standard Error	0.005151391	Standard Error	0.387084894	Standard Error	7.492388692
Median	0.538	Median	5	Median	330
Mode	0.538	Mode	24	Mode	666
Standard Deviation	0.115877676	Standard Deviation	8.707259384	Standard Deviation	168.5371161
Sample Variance	0.013427636	Sample Variance	75.81636598	Sample Variance	28404.75949
	-		-		-
Kurtosis	0.064667133	Kurtosis	0.867231994	Kurtosis	1.142407992
Skewness	0.729307923	Skewness	1.004814648	Skewness	0.669955942
Range	0.486	Range	23	Range	524
Minimum	0.385	Minimum	1	Minimum	187
Maximum	0.871	Maximum	24	Maximum	711
Sum	280.6757	Sum	4832	Sum	206568
Count	506	Count	506	Count	506

PTRATIO		AVG_ROOM	
Mean	18.4555336	Mean	6.284634387
Standard Error	0.096243568	Standard Error	0.031235142
Median	19.05	Median	6.2085
Mode	20.2	Mode	5.713
Standard Deviation	2.164945524	Standard Deviation	0.702617143
Sample Variance	4.686989121	Sample Variance	0.49367085
Kurtosis	-0.285091383	Kurtosis	1.891500366
Skewness	-0.802324927	Skewness	0.403612133
Range	9.4	Range	5.219
Minimum	12.6	Minimum	3.561
Maximum	22	Maximum	8.78
Sum	9338.5	Sum	3180.025
Count	506	Count	506

LSTAT		AVG_PRICE	
Mean	12.65306324	Mean	22.53281
Standard Error	0.317458906	Standard Error	0.408861
Median	11.36	Median	21.2
Mode	8.05	Mode	50
Standard Deviation	7.141061511	Standard Deviation	9.197104
Sample Variance	50.99475951	Sample Variance	84.58672
Kurtosis	0.493239517	Kurtosis	1.495197
Skewness	0.906460094	Skewness	1.108098
Range	36.24	Range	45
Minimum	1.73	Minimum	5
Maximum	37.97	Maximum	50
Sum	6402.45	Sum	11401.6
Count	506	Count	506

- Using the data analysis tool we can understand the characteristic's of each variable
Like Mean,Median,Mode,Maximum,Minmum etc.
- In this data we can observe like “Tax” has highest mean,standard deviation.
- “Avg room” has more positive Kurtosis around(1.89) value.
- “Avg price” has more positive Skewness arouound (1.108) value.

2) Plot a histogram of the Avg_Price variable. What do you infer?



- In the above Histogram we can observe that house price around 21 to 25 has highest frequency.
- There are relatively fewer houses with extremely high above 40 and extremely low price below 10.

3) Compute the covariance matrix. Share your observations

	CRIME_ RATE	AGE	INDUS	NOX	DISTAN CE	TAX	PTRATI O	AVG_R OOM	LSTAT	AVG_P RICE
CRIME_ RATE	8.516147 873									
AGE	0.562915 215	790.792 4728								
INDUS	0.110215 175	124.267 8282	46.9714 2974							
NOX	0.000625 308	2.38121 1931	0.60587 3943	0.01340 1099						
DISTAN CE	0.229860 488	111.549 9555	35.4797 1449	0.61571 0224	75.6665 3127					
TAX	8.229322 439	2397.94 1723	831.713 3331	13.0205 0236	1333.11 6741	28348.6 236				
PTRATI O	0.068168 906	15.9054 2545	5.68085 4782	0.04730 3654	8.74340 249	167.820 8221	4.67772 6296			
AVG_RO OM	0.056117 778	4.74253 803	1.88422 5427	0.02455 4826	1.28127 7391	34.5151 0104	0.53969 4518	0.49269 5216		
LSTAT	0.882680 362	120.838 4405	29.5218 1125	0.48797 9871	30.3253 9213	653.420 6174	5.77130 0243	3.07365 4967	50.8939 7935	
AVG_PR ICE	1.162012 24	97.3961 5288	30.4605 0499	0.45451 2407	30.5008 3035	724.820 4284	10.0906 7561	4.48456 5552	48.3517 9219	84.4195 6

- The Diagonal elements of the variance matrix represents the variance of individual variable ,these variables indicate how much the values of each variable deviate from the mean,for example of Crime_rate is 8.52 which is variance of the age around 790.79.
- The off diagonal elements represents the covariance between pairs of variables Covariance measures how two variables vary together.positive Covariance indicates that when one variable increase,the other tends to increases as well.negative covariance indicates that when one variable increases.on other tends to decreases.
- For the instance there is positive Covariance between Crime_rate and Age age increases and also the Crime rate increases.

4) Create a correlation matrix of all the variables

a) Which are the top 3 positively correlated pairs

b) Which are the top 3 negatively correlated pairs

	CRIME_ RATE	AGE	INDUS	NOX	DISTAN CE	TAX	PTRATI O	AVG_R OOM	LSTAT	AVG_P RICE
CRIME_ RATE	1									
AGE	0.006859 463	1								
INDUS	0.005510 651	0.64477 8511	1							
NOX	0.001850 982	0.73147 0104	0.76365 1447	1						
DISTAN CE	0.009055 049	0.45602 2452	0.59512 9275	0.61144 0563	1					
TAX	0.016748 522	0.50645 5594	0.72076 018	0.66802 32	0.91022 8189	1				
PTRATI O	0.010800 586	0.26151 5012	0.38324 7556	0.18893 2677	0.46474 1179	0.46085 3035	1			
AVG_RO OM	0.027396 16	0.24026 4931	0.39167 5853	0.30218 8188	0.20984 6668	0.29204 7833	0.35550 1495	1		
LSTAT	0.042398 321	0.60233 8529	0.60379 9716	0.59087 8921	0.48867 6335	0.54399 3412	0.37404 4317	0.61380 8272	1	
AVG_PR ICE	0.043337 871	0.37695 4565	0.48372 516	0.42732 0772	0.38162 6231	0.46853 5934	0.50778 6686	0.69535 9947	0.73766 2726	1

Top 3 positive Correlated pairs are

- Distance and Tax -> 0.910228
- Indus and Nox -> 0.763651
- Age and Nox -> 0.731470

Top 3 negative Corelated pairs are

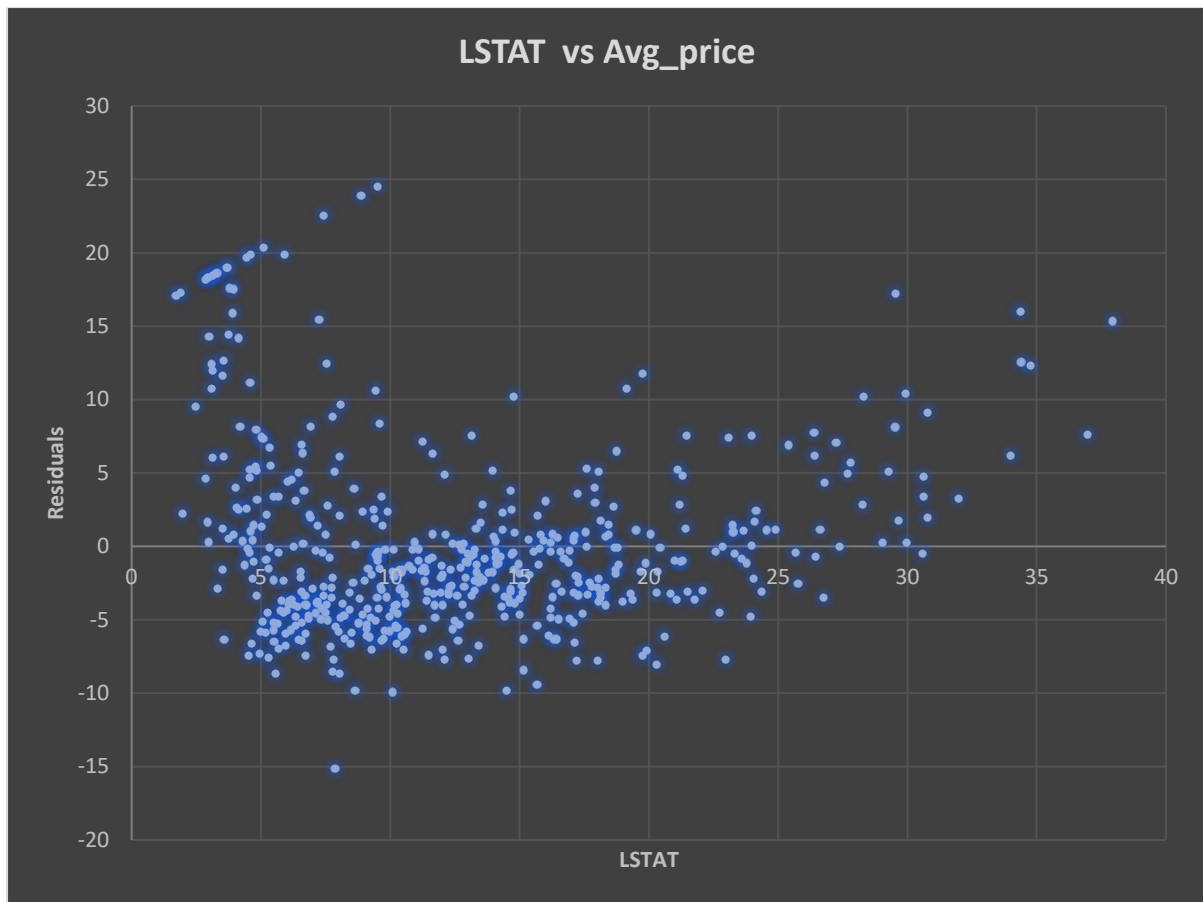
- Lstat and Avg_price -> -0.73766
- Avg_room and Lstat -> -61381
- Ptratio and Avg_price -> -0.507786.

5) Build an initial regression model with AVG_PRICE as ‘y’ (Dependent variable) and LSTAT variable as Independent Variable. Generate the residual plot.

a) What do you infer from the Regression Summary output in terms of variance explained, coefficient value, Intercept, and the Residual plot?

b) Is LSTAT variable significant for the analysis based on your model?

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.737662726							
R Square	0.544146298							
Adjusted R Square	0.543241826							
Standard Error	6.215760405							
Observations	506							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	1	23243.914	23243.91	601.6178711	5.08E-88			
Residual	504	19472.38142	38.63568					
Total	505	42716.29542						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	34.55384088	0.562627355	61.41515	3.7431E-236	33.44846	35.65922	33.44845704	35.65922
LSTAT	-0.950049354	0.038733416	-24.5279	5.0811E-88	-1.02615	-0.87395	-1.0261482	-0.87395



5A.

- Variance (R square) value is around 0.5441, approximately 54.41% , R value tells us how independent variable influences the dependent variable.
- Coefficient value LSTAT IS (-0.95004) it shows us negative Coefficient value.
- Intercept value is 34.5538 ,when Avg_price and LSTAT value is equal to 0.

5B.

- The LSTAT Variable is Significant based on P value ,based on above data the p value is around (5.0811E) which is less than 0.05 p value so we can conclude this model is Significant.

6) Build a new Regression model including LSTAT and AVG_ROOM together as Independent variables and AVG_PRICE as dependent variable

a) Write the Regression equation. If a new house in this locality has 7 rooms (on an average) and has a value of 20 for L-STAT, then what will be the value of AVG_PRICE? How does it compare to the company quoting a value of 30000 USD for this locality? Is the company Overcharging/ Undercharging?

b) Is the performance of this model better than the previous model you built in Question 5? Compare in terms of adjusted R-square and explain.

6A.

The Avg_Price = $-1.3583 + 5.0948 \cdot \text{Avg_room} - 0.6424 \cdot \text{Lstat}$

Lstat value is 20

Avg_room are 7.

$$\begin{aligned}\text{Avg_price} &= -1.3583 + 5.0948 \cdot 7 - 0.6424 \cdot 20 \\ &= 21.4573\end{aligned}$$

Avg_price is 21.4573 is less than 30000 USD ,The company is Overcharging.

6B

The R value is 0.5432 when we consider single independent variable LSTAT.

Adjusted R Square = 0.6371.

When we consider two independent variables like LSTAT And Avgroom the Radjusted value is more compare to the previous model

7) Build another Regression model with all variables where AVG_PRICE alone be the Dependent Variable and all the other variables are independent. Interpret the output in terms of adjusted Rsquare, coefficient and Intercept values. Explain the significance of each independent variable with respect to AVG_PRICE

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.832978824							
R Square	0.69385372							
Adjusted R	0.688298647							
Standard E	5.1347635							
Observations	506							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	9	29638.8605	3293.206722	124.9045049	1.9328E-121			
Residual	496	13077.43492	26.3657962					
Total	505	42716.29542						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	29.24131526	4.817125596	6.070282926	2.53978E-09	19.77682784	38.70580267	19.77682784	38.70580267
CRIME_RATE	0.048725141	0.078418647	0.621346369	0.534657201	-0.105348544	0.202798827	-0.105348544	0.202798827
AGE	0.032770689	0.013097814	2.501996817	0.012670437	0.00703665	0.058504728	0.00703665	0.058504728
INDUS	0.130551399	0.063117334	2.068392165	0.03912086	0.006541094	0.254561704	0.006541094	0.254561704
NOX	-10.3211828	3.894036256	-2.650510195	0.008293859	-17.97202279	-2.670342809	-17.97202279	-2.670342809
DISTANCE	0.261093575	0.067947067	3.842602576	0.000137546	0.127594012	0.394593138	0.127594012	0.394593138
TAX	-0.01440119	0.003905158	-3.687736063	0.000251247	-0.022073881	-0.0067285	-0.022073881	-0.0067285
PTRATIO	-1.074305348	0.133601722	-8.041104061	6.58642E-15	-1.336800438	-0.811810259	-1.336800438	-0.811810259
AVG_ROOMS	4.125409152	0.442758999	9.317504929	3.89287E-19	3.255494742	4.995323561	3.255494742	4.995323561
LSTAT	-0.603486589	0.053081161	-11.36912937	8.91071E-27	-0.70777824	-0.499194938	-0.70777824	-0.499194938

- **CRIME RATE** is not Significant variable because of pvalue which is greater then 0.05 value.
- **AGE,INDUS,NOX,DISTANCE,TAX,PTRATIO,AVG_ROOMS,LSTAT** Are Significant variables with respective to the p values which are less 0.05.

8) Pick out only the significant variables from the previous question. Make another instance of the Regression model using only the significant variables you just picked and answer the questions below

a) Interpret the output of this model.

b) Compare the adjusted R-square value of this model with the model in the previous question, which model performs better according to the value of adjusted R-square?

c) Sort the values of the Coefficients in ascending order. What will happen to the average price if the value of NOX is more in a locality in this town?

d) Write the regression equation from this model.

8a.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.832835773							
R Square	0.693615426							
Adjusted R Square	0.688683682							
Standard Error	5.131591113							
Observations	506							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	8	29628.68142	3703.585178	140.6430411	1.911E-122			
Residual	497	13087.61399	26.33322735					
Total	505	42716.29542						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	29.42847349	4.804728624	6.124898157	1.84597E-09	19.98838959	38.8685574	19.98838959	38.8685574
AGE	0.03293496	0.013087055	2.516605952	0.012162875	0.007222187	0.058647734	0.007222187	0.058647734
INDUS	0.130710007	0.063077823	2.072202264	0.038761669	0.006777942	0.254642071	0.006777942	0.254642071
NOX	-10.27270508	3.890849222	-2.640221837	0.008545718	-17.9172457	-2.628164466	-17.9172457	-2.628164466
DISTANCE	0.261506423	0.067901841	3.851242024	0.000132887	0.128096375	0.394916471	0.128096375	0.394916471
TAX	-0.014452345	0.003901877	-3.703946406	0.000236072	-0.022118553	-0.006786137	-0.022118553	-0.006786137
PTRATIO	-1.071702473	0.133453529	-8.030529271	7.08251E-15	-1.333905109	-0.809499836	-1.333905109	-0.809499836
AVG_ROOM	4.125468959	0.44248544	9.323400461	3.68969E-19	3.256096304	4.994841615	3.256096304	4.994841615
LSTAT	-0.605159282	0.0529801	-11.42238841	5.41844E-27	-0.70925186	-0.501066704	-0.70925186	-0.501066704

- R value is around 0.6936154 which is around approximately 69.36 % which is Good model against the dependent variable (Avg_price) and all remaining independent variables.

8B.

- When we compare two adjusted r values of 7 and 8 model.

The r value of 7 model is around 0.6982 and r value of 8 model is 0.6986 there is slightly difference value,

So that we should consider the 8 model that is 69.86.

8c

NOX	-10.27270508
PTRATIO	-1.071702473
LSTAT	-0.605159282
TAX	-0.014452345
AGE	0.03293496
INDUS	0.130710007
DISTANCE	0.261506423
AVG_ROOM	4.125468959
Intercept	29.42847349

- If the value of Nox value is more in locality the avg_price is decreases because measures of nitrogen oxide emissions.
- Higher the value of nitrogen oxide emissions the property value is lower.

8D

$$Y = B_0 + B_1X_1 + B_2X_2 + \dots + B_nX_n$$

$$\text{Avg_price} = 29.42 + 0.0329 + 0.1307 + (-1.02727) + (-0.60515).$$